NGOs, Requirements, and Evaluation Criteria

Need

Balmy Beach Community Daycare needs a fast, reliable way to apply identification wristbands to all attending children on the morning of a field trip.

Goals

1. Design for **Efficiency:** The design should enable efficient wristband preparation and application during the morning arrival period.
2. Design for **Reliability:** Support constant and reliable wristband application across varying sizes and staffing conditions.
3. Design for **Usability:** The wristband-related design should be usable for all day care children.
4. Design for **Safety:** Ensure safety of everyone involved throughout wristband preparation and application.

Goals	Objective 1	Objective 2	Objective 3
Design for Efficiency	Design setup shall be ≤ 1 minute of active staff time per field trip (for up to 90 wristbands)	The design shall allow wristband application to take < 34 seconds and less than 5 steps.	The design must not take more than 10 minutes to prepare.
Design for Reliability	Enable a repeatable process that eliminates 100% of the need for staff to manually verify that each child is wearing a wristband.	Ensure that the wristband application method or design maintains safe and reliable operation for > 5 years.	
Design for Usability	The wristband application process takes at most 3 steps per child for day care children or staff to perform	The design must be usable by children and staff spanning the 5th-95th percentile hand size.	
Design for Safety	The design shall meet the UL 1439 Sharp Edge Test standard.	Eliminate exposure to pinch and entrapment hazards during wristband preparation and application.	The maximum force applied to or by the wrist during application should be less than 10 N.

Goal 1**: Efficiency**

Requirements	**Verification**	**Justification**	**Evaluation Criteria**
The design shall require **≤ 60 seconds** of active staff time to set up before distribution/wristband application begins.	Timed proxy test: simulate the set up process for the design, including placing the wristband in the design (if necessary), and ensuring the design is ready for use without further adjustments.	Setup time is limited to ≤ 1 minute because this limit classifies the device as a One Touch Exchange of Die (OTED) according to Kaizen Institute, which means that this setup time prevents adding a meaningful new task to staff workload. Since the setup cannot meaningfully disrupt workflow, staff will have more time to focus on other important tasks.	Set up time (**seconds**): shorter is better
The wristband application process shall require less than or equal to **5 steps**.	Procedure analysis: analyze each step of the wristband application process, and count each step required to attach the wristband, and remove the excess.	The current wristband application process involves over 6 steps. Therefore, any possible design must have less steps in order to improve the current process.	**Number of steps**: less is better
The design shall allow a wristband to be applied in ≤ **34 seconds**.	Timed proxy test: simulate applying a wristband on a cooperative individual and record the time. Take multiple trials to ensure consistency through all trials.	The wristband application process was selected to take less than 34 seconds based on a Methods-Time Measurement (MTM) analysis, which predicts that applying an adjustable snap-closure ID wristband to a cooperative individual under normal conditions requires approximately 34 seconds. This value	Application time per wristband (**seconds**): shorter is better

		represents an ideal baseline time assuming minimal resistance or interruption.	
The design shall require **≤ 5-10 minutes** to prepare 90 wristbands before set-up.	Timed proxy test: measure the time to prepare 10 wristbands, then multiply by 9 for 90 wristbands to get an estimation for the preparation time.	In the morning, staff arrive 5 minutes early to prepare for their shift. Additionally, early learning specialists say that it is important to minimize transition times such as the time spent preparing the children before applying the wristbands. Therefore, a range of 5-10 minutes is appropriate.	Total preparation time (**minutes**): shorter is better
The design shall contain identification information of the Daycare thats legible as per ISO/TR 22411 section 6.2.7-6.2.9	Visual inspection for legibility and contrast.	Standardizes clarity for diverse visual abilities to ensure rapid identification in emergencies.	Legibility: Higher contrast/font size is better.

Goal 2: **Reliability**

Requirements	Verification	Justification	Evaluation Criteria
The design shall keep the verification time of each kid to be in ≤ **5 seconds**.	Do a proxy field trip containing 50 students using the wristband application system. Ask staff to perform a verification pass to verify that all kids are wearing the wristband. Record the total time during the process and divide the time to the number of kids to get the verification time per kid.	Staff have to manage multiple tasks during preparations for the fieldtrips, so if verification that each child is wearing a wristband requires too much time, it will increase the work load and decrease the supervision capability. Therefore, the repeated verification should be quick and effortless.	Calculate the verification time per child. The time has to be ≤ **5 seconds**: shorter is better.
The design shall stay reliable after being used for **5 years**.	It is unrealistic to observe whether the wristbands remain functional after a five-year period, so do a proxy test of an exaggerated damaging process to one wristband, record the total time being used until the wristband cannot be used.	School boards and childcare facilities plan replacements for durable assets on a five to seven-year cycle. The Toronto District Board's Equipment Replacement Plan treats programme equipment with a five-year life expectancy for budgeting purposes. Therefore, a five-year service life is consistent with equipment expectations in licensed childcare and educational environments.	Record the time spent before the wristband is not usable (**minutes**): shorter is better.
The wristband shall be capable of remaining securely attached during normal daycare	Wear simulation test with repeated motion and handling cycles	From the meeting with stakeholders, the community mentioned that some kids might leave on multiple activities in one week, so they hoped that	**Retention** time before failure: longer is better

activities for up to **1 week.**		the wristband could remain functional for one week.	
The design shall meet the **5 requirements** of waterproof watches outlined in ISE ISO 22810:2010	Outlined in detail in the standard	If the wristband has mechanical components, it is analogous to a watch - likely with lower complexity. The ISO 22810:2010 standard defines water resistant watches	**Pass/fail:** Better if it meets a higher level of the standard.

Goal 3: **Usability**

Requirements	Verification	Justification	Evaluation Criteria
The design shall ensure that each application can be completed using **single simple actions** (e.g. place, fasten, remove access)	Usability observation during user testing	Students in Balmy Beach Daycare have the ages ranging from 4 to 12, so young kids can only understand and follow some simple steps.	**Complexity** of actions per step: simpler actions are better
The design shall allow the wristband application process to be **understood** and **executed** by users **aged 4+**	Usability testing or instruction-following task with representative users.	The youngest kids in the daycare center are the age of 4.	**Learning time** for process: shorter is better
The design shall accommodate users within the **5th–95th percentile** anthropometric hand size range.	Anthropometric comparison using ISO 7250-3 hand size data.	Designs for the full 0th-100th percentile is impractical and will negatively impact the usability, precision and comfort. 5th-95th percentile provides an appropriate balance.	**Percentage** of users accommodated: higher is better
The wristband shall allow a clearance of **10–20 mm** between the wristband and the child’s wrist.	Measure clearance using calipers or finger-clearance test after application.	This length is equivalent to the two-finger spacing guideline, which could prevent excessive compression while maintaining retention.	Wrist **clearance**: adequate clearance without looseness

The wristband shall weigh **≤30g.**	Weigh the entire band or do mathematical calculations with predicted material contents.	Heavy watches cause strain on wrists, which can cause long term effects in developing children. Similarly, wristbands that are 20-60g are considered comfortable for everyday wear for adults. Through looking at a comprehensive data set of children's wrist watches, 30g was typically the highest they weigh comfortably. A 5th percentile 2.5 year old is 10kg, 0.25% of that is 25g.	**Mass**: lower is better.
Attachments to the wristband shall be within **25-28mm** diameter and between **8-10mm** height.	Measure the dimensions of wristband components that are not the band	Older children are capable of adapting to a design better than younger, so we will approach from the perspective of the smallest and youngest children in the scope. A 5th percentile 2.5 year old's wrist breadth is 28.4mm and width is 21.5mm. Finger pads are 8mm so any smaller than this would cause difficulty gripping the device. Attachments must sit comfortably on the wrist and not be overly large. Additionally, in-hand manipulation occurs most efficiently when the object is 50-75% of the hand width. A 5th percentile 2.5 year old has a hand width of 50.2mm so the lowest ideal is 25.1mm.	**Size:** Close to 26.5mm diameter and 9mm height the better.

Goal 4: **Safety**

Requirements	Verification	Justification	Evaluation Criteria
Any wristband and application mechanism shall have **all edge**s fulfill the **UL1439 Sharp Edge Test standard.**	Proxy test: create a probe with 12.7mm diameter and wrap the end with electrical tape to act as the skin. Drag it over the prototype edges for 50mm with a force of 6.67N or 0.68 kg, measured via a zeroed weight scale below. If tape is not ruptured it passes.	Provides a standardized, repeatable and Canadian industry-recognized method for evaluating the safety of product edges. If a maximum of one layer of the UL sensing tape is cut, this confirms that all contact surfaces are sufficiently smooth and rounded to avoid cuts or abrasions during normal handling	**Force** before rupture (N): lower is better. **Deformation** of the tape (mm): lower is better. Ratio of **rounded edges** to sharp corners: higher is better.
Any wristband application mechanism shall have **0 accessible pinch points and entrapment hazards** during any step in the usage process.	Proxy Test: No moving parts shall be accessible to a 8mm diameter probe - representing a child's finger	The Canadian Centre for Occupational Health and Safety identifies moving parts or machinery as a category for physical hazards, which includes pinching, entanglement and trapping hazards that can cause injury if a body part is in contact or is caught between them.	Gap sizes of **<5mm,** or **>15mm,** the better. **Speed** of moving parts (m/s): lower is better.
Coverings of mechanical points and other hazards must withstand **66.8N** of tension and **133.4N** of compression **force**.	For high-fidelity prototypes, forces can be applied directly to determine pass/fail. Alternatively, mathematical prototypes based off of scientific principles can be used during early stages of design development.	According to ASTM F963, components must withstand 66.8 N of force of tension in order to be non-detachable for children ages 3-14. Additionally, it needs to withstand 133.4N of compression to prevent failure under pressure. This ensures that any coverings of potential hazards are effective.	**Force** withstood: higher is better. **Cycles of testing** without failure: higher is better.

Any wristband application process shall apply a **force** of no more than **13.5N** to users wrists.	For high-fidelity prototypes, forces can be applied directly to determine pass/fail. Alternatively, mathematical prototypes based off of scientific principles can be used during early stages of design development.	The Pain Pressure Threshold found children aged 4-14 feel pain at a force of 18N on average across a 1cm area. This max provides a buffer before that occurs.	**Force** applied to wrist: lower is better. **Distribution** of force: higher is better.
Any manual steps in the application process shall require no more than **22N** of **force** to complete.	For high-fidelity prototypes, forces can be applied directly to determine pass/fail. Alternatively, mathematical prototypes based off of scientific principles can be used during early stages of design development.	Force limits from accessibility and human-factors standards specify that hardware intended for one-handed operation without tight grasping or twisting should require no more than 22 N of force.	**Force** required by wrist: lower is better. **Duration** of applied force: lower is better.
The wristband shall never apply a pressure **≥32mmHg** to the wearer.	Tension tests	The average pressure in a human arteriolar limb is 32mmHg. Any pressure greater than this will exceed the internal pressure and cause it to collapse, thus providing a safe upper bound.	**Pressure** applied to wrist: lower is better.
The wristband shall break or release at **67N** of tension force.	Test force to failure with a force gauge.	The ASTM F2923-20 Standard for Children's Jewelry indicates neck jewelry must break under 15 pounds of tension to prevent strangulation. This works as a safety factor for limb safety as applied.	Failure **Force:** lower is better.
The materials of the wristband shall be **100%** child safe and non-irritating,	Wear tests	The wristband is considered a surface device with prolonged contact. These standards ensure there is no	Amount of **potential irritants**: lower is better.

fulfilling ISO 10993-5 & ISO 10993-10.		cytotoxicity, irritation, or skin sensitivity	